

Models	b5M	b10M	b15M	b20M	ba20M	bb20M
W/o TM	41.11	41.40	42.53	43.19	41.47	42.53
Directly	41.77	42.45	44.27	44.26	41.87	43.75
Comma	41.29	41.88	43.79	43.99	41.18	43.36
Semico	42.27	42.45	44.64	45.13	42.12	44.28
Conjunc	41.48	42.04	43.87	44.33	41.54	43.09
Parenth	41.68	42.63	44.49	44.53	41.93	43.85
Max Δ	1.16	1.23	2.11	1.94	0.65	1.75

Table 7: Experimental results on UNPC En-Zh test set with different NMT models, including four transformer big models trained on 5 million, 10 million, 15 million, and 20 million training data (denoted as b5M, b10M, b15M, b20M, respectively), and a transformer base and a bigger model trained on 20 million training data (denoted as ba20M and bb20M, respectively), max Δ denotes the maximum improvement comparing to decoding without TM.

From the experimental results in Table 7, we have the observations below.

For the same model architecture, with the increase of training data scale, the translation ability of the model is getting stronger, and the BLEU improvement of our method is also higher compared with the baseline, as the maximum BLEU score improvement of the models b15M and b20M are higher than that of b5M and b10M. In addition, for the models trained on the same training data, the ba20M model is “weaker” than the b20M and bb20M models, and the maximum BLEU score improvement is also lower than the latter two models. We can find a similar phenomenon if we look back to Table 4 and Table 6. The dataset for UNPC En-Zh is the same in these two experiments, and the NMT model of the commercial NMT system is much “strong” than the NMT model used in table 4, and the maximum improvement of the former is an 2.94 BLEU points, whereas the latter’s is 0.89 BLEU points. From these results, we conclude that the sentence-level TM approach of our method can further improve strong baselines, and the “stronger” the NMT model, the greater the improvement will be.

TM Similarity

As our method obtain useful information from a single TM sentence pair, the translation result is influenced by the similarity of the retrieved TM. TMs with high similarity provide more useful information for the translation, while less similar ones may introduce noise to decoding. In this experiment, we explore the similarity threshold that a TM can help or not. We decoded the UNPC En-Zh test set using NiuTrans WMT20 En-Zh model (?). Specifically, the test set is divided into various portions based on the similarity score, and each portion of the test set is decoded individually both with the sentence-level TM approach and baseline (without TM).

From the experimental results in Figure 3, we can find that the BLEU scores of our method are lower than the baseline when the similarity score is lower than 0.6; when it is between 0.6 and 0.8, some of our concatenation methods perform better than the baseline with a marginal advantage;

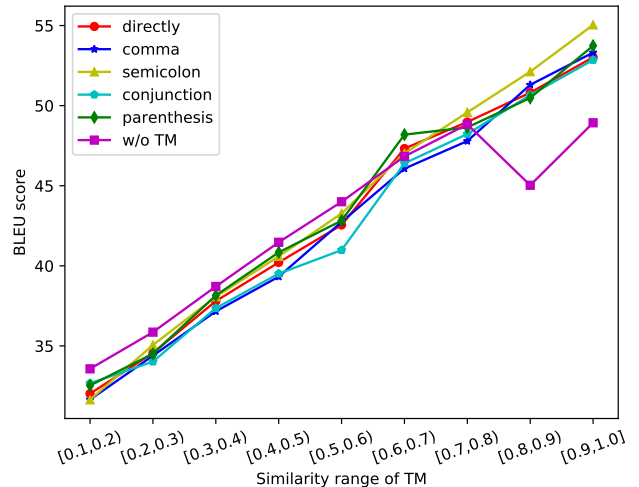


Figure 3: BLEU scores on different similarity ranges.

when it is higher than 0.8, all of the concatenation methods outperform the baseline significantly, with a maximum improvement of 7.09 and 6.11 BLEU points, respectively. Therefore, the FMS threshold for sentence-level TM of the NiuTrans WMT20 En-Zh model is 0.8.

The threshold is determined by the “strength” of the NMT model that “strong” models are more robust to obtaining useful information and avoiding noises introduced by less similar TMs. Thus, “strong” models have lower FMS thresholds. In practice, the threshold can be used to decide whether to apply TM for decoding.

Sentence Length

In this section, we investigate the impact of input sentence length on translation. To avoid the TM similarity influencing the experimental results, we used the test set itself as the retrieved TM, which means that the TMs are 100% similar to the input sentences. We split the test set into groups according to the length of the input sentence, and uniformly chose 170 sentences from each group as the test set (the minimum sentence number of the original groups is 171). The experimental results are given in Figure 4. From the experimental results, we can find a sharp tendency that the performance of our proposed method decreases and eventually be comparable to the baseline as the sentence length increases. The performance of the baseline, however, tends to be steady. This is also determined by the initial design of our method. For an input sentence, we employ an NMT model that has been trained on a sentence-level dataset. As we concatenate the source TM with the input sentence before feeding them into the NMT model, the length of the input for the model encoder will be doubled. In this way, the whole sentence length of a lengthy sentence and its corresponding TM will deviate from the training model’s sentence length distribution. This is the main cause of our method’s performance in the figure degrading on lengthy sentences. Therefore, our method can not handle long sentences well even though a highly similar sentence is retrieved. After calculation, we find that the average sentence length of the UNPC En-Zh test set and homemade En-Zh test set in Table 4 are 29.58 and 10.99. This is why

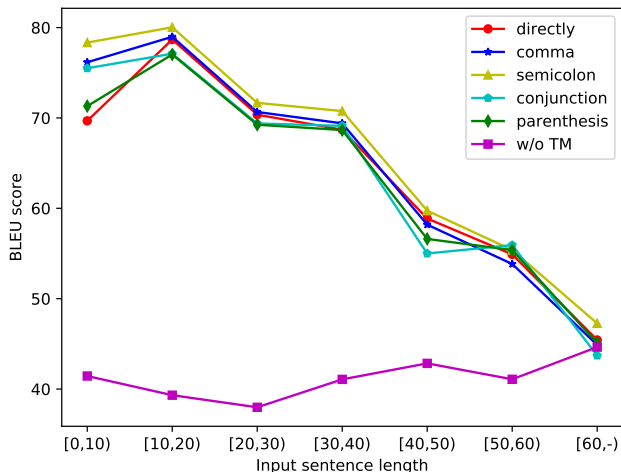


Figure 4: input sentence length impact on BLEU.

the latter can improve the translation more significantly than the former even though there are fewer sentences with high similarity than there are in the former.

Related Work

Many studies have been conducted in recent years to enhance MT quality using TMs. With the emergence of NMT, the MT community is seeing an increasing interest in TM research. There are mainly two research lines for TM integration into NMT: constraining the decoding process with TM and using TM to train a more powerful NMT model. The main idea of the first research line is to increase the generation probability of some target words based on TM. ? constrained the decoding process by increasing the generation possibility of the target words which are in the aligned slices extracted from the TM. Following this work, ? added positional information for words in the TM slices. ? and ? embedded the retrieved TM information into the NMT model by fine-tuning the NMT model with TMs before translating the input sentence. Instead of incorporating text level TM, k NN-MT retrieved TMs from dense vectors (?). First, they created a key-value datastore from the TM database, where the key is the translation context vector of each time step, and the value is the true target token. In the inference time, k NN-MT interpolates the generation probability of the NMT model and retrieved similar target distribution from that datastore at each time step. Following k NN-MT, several researches optimized k NN-MT from different perspectives. ? accelerated the inference process by narrowing the search range, instead of searching from the entire data store. By introducing a lightweight meta-k network, ? dynamically determines how many neighbors should be introduced. ? further accelerated k NN-MT inference by constraining search space when constructing the data store. Instead of retrieving a single token, ? retrieved chunks of tokens from the data store to speed up k NN-MT. The second research line aims to train the generation model to learn how to deal with the retrieved TMs. ??? used a data augmentation way to

concatenate the retrieved TM with input sentence during training. One sub-method of ? generated the target prefix in a forced way when decoding, while they need to train the NMT model with a similar concatenation pattern first. Some researches modified the NMT model architecture for better TM integration. ? and ? introduced a gating mechanism module to control the signal from the retrieved TM. ? designed an additional transformer encoder to integrate the target sentence of TM through the attention mechanism. In ?, the retrieved multiple TMs are compressed into a graph structure for speed up and space savings and then are integrated into the model via the attention mechanism. ? proposed a lightweight method to incorporate the target sentence of retrieved TM in an extra attention module. Unlike all of the above methods, ? proposed a method to incorporate monolingual TM into NMT, and the target sentence retriever and NMT model are trained jointly.

Conclusion and Future Work

In this paper, we propose a simple but effective method to incorporate TM into NMT decoding without modifying the pre-trained NMT model. Specifically, we treat the retrieved TMs as prompts for the translation of the input sentence by concatenating the source TM with the input sentence and generating the target token in a forced way. Experiments on the TM specialized translation task, domain adaptation task, and implementation on commercial MT system verify the effectiveness of our method. Our method is easy to implement and can be applied to customize a TM-incorporated machine translation system for TM data on the user side. Our method in this paper suffers from TM sentences with low similarity scores and long sentences. In the future, we will investigate more effective methods to alleviate the drawbacks of our methods in low similarity TM and long sentence translation situations.

Acknowledgments

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